

Lynbrook Math Summer Camp 2026

N5 Modular Arithmetic

Lynbrook Math Club

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N5.1 Lecture

Can a perfect square have remainder 2 when divided by 4?

Or consider the much larger-looking question:

What is the units digit of 7^{2026} ?

At first, these problems seem unrelated. However, both involve finding remainders of a quantity when divided by some number (4 and 10 respectively). Modular arithmetic gives us a precise way to describe and work with these cycles.

Congruence

Definition N5.1.1 (Congruence). Let a and b be integers, and let m be a positive integer. We say that a is **congruent** to b modulo m , written

$$a \equiv b \pmod{m},$$

if a and b leave the same remainder when divided by m .

Equivalently,

$$a \equiv b \pmod{m} \iff m \mid (a - b).$$

For example,

$$17 \equiv 5 \pmod{12}$$

because both numbers leave remainder 5 when divided by 12. So,

$$12 \mid (17 - 5).$$

Negative residues are often useful as well. For example,

$$20 \equiv -1 \pmod{7}$$

because

$$20 - (-1) = 21$$

is divisible by 7.

Remark N5.1.2. Every integer is congruent modulo m to exactly one of

$$0, 1, 2, \dots, m-1.$$

These are called the **least nonnegative residues modulo m** .

Arithmetic with Congruences

Congruences behave well under addition, subtraction, and multiplication.

Claim N5.1.3 — If

$$a \equiv b \pmod{m} \quad \text{and} \quad c \equiv d \pmod{m},$$

then

$$a + c \equiv b + d \pmod{m},$$

$$a - c \equiv b - d \pmod{m},$$

and

$$ac \equiv bd \pmod{m}.$$

Proof. Since

$$m \mid (a - b) \quad \text{and} \quad m \mid (c - d),$$

we have

$$m \mid ((a - b) + (c - d)),$$

so

$$a + c \equiv b + d \pmod{m}.$$

The subtraction case is similar.

For multiplication,

$$ac - bd = a(c - d) + d(a - b).$$

Both terms on the right are divisible by m , so

$$m \mid (ac - bd).$$

Therefore,

$$ac \equiv bd \pmod{m}.$$

□

Example 4 (Addition)

Find the remainder when

$$123456 + 789012$$

is divided by 9.

Recall properties of division by 9 from the divisibility class. We have

$$123456 \equiv 3 \pmod{9}$$

and

$$789012 \equiv 0 \pmod{9}.$$

Therefore,

$$123456 + 789012 \equiv 3 + 0 \equiv 3 \pmod{9}.$$

So the remainder is

$$\boxed{3}.$$

The example for subtraction is similar.

Example 5 (Multiplication)

Find the remainder when

$$1234 \cdot 5678$$

is divided by 10.

Reduce each factor modulo 10:

$$1234 \equiv 4 \pmod{10} \quad \text{and} \quad 5678 \equiv 8 \pmod{10}.$$

Therefore,

$$1234 \cdot 5678 \equiv 4 \cdot 8 = 32 \equiv 2 \pmod{10}.$$

So the remainder is

$$\boxed{2}.$$

Repeated multiplication gives the following useful consequence.

Claim N5.1.6 — If

$$a \equiv b \pmod{m},$$

then for every positive integer n ,

$$a^n \equiv b^n \pmod{m}.$$

More generally, if $P(x)$ is a polynomial with integer coefficients and

$$a \equiv b \pmod{m},$$

then

$$P(a) \equiv P(b) \pmod{m}.$$

Why Division Is Different

Although addition, subtraction, and multiplication work naturally with congruences, division requires care.

From

$$ac \equiv bc \pmod{m},$$

we cannot always conclude that

$$a \equiv b \pmod{m}.$$

For example,

$$2 \cdot 1 \equiv 2 \cdot 4 \pmod{6}$$

because

$$2 \equiv 8 \pmod{6}.$$

However,

$$1 \not\equiv 4 \pmod{6}.$$

The problem is that the factor 2 shares a common factor with the modulus 6.

Claim N5.1.7 (Cancellation Rule) — If

$$ac \equiv bc \pmod{m}$$

and

$$\gcd(c, m) = 1,$$

then

$$a \equiv b \pmod{m}.$$

Proof. The congruence

$$ac \equiv bc \pmod{m}$$

means that

$$m \mid c(a - b).$$

Since

$$\gcd(c, m) = 1,$$

Euclid's lemma implies that

$$m \mid (a - b).$$

Therefore,

$$a \equiv b \pmod{m}.$$

□

Example 8

Solve

$$4x \equiv 12 \pmod{9}.$$

Since

$$\gcd(4, 9) = 1,$$

we may cancel 4 after observing that $12 \equiv 3 \pmod{9}$. Equivalently, since

$$4^{-1} \equiv 7 \pmod{9},$$

we obtain

$$x \equiv 7 \cdot 3 = 21 \equiv 3 \pmod{9}.$$

Therefore,

$$\boxed{x \equiv 3 \pmod{9}}.$$

Modular Inverses

Definition N5.1.9 (Modular Inverse). An integer b is called a **multiplicative inverse** of a modulo m if

$$ab \equiv 1 \pmod{m}.$$

Here, b is often written as a^{-1} , read "a inverse".

For example,

$$3 \cdot 5 = 15 \equiv 1 \pmod{7},$$

so 5 is an inverse of 3 modulo 7.

The existence of inverses is closely connected to Bézout's identity from the GCD and LCM lesson.

Theorem N5.1.10

An integer a has a multiplicative inverse modulo m if and only if

$$\gcd(a, m) = 1.$$

Proof. Suppose first that a has an inverse modulo m . Then there exists an integer x such that

$$ax \equiv 1 \pmod{m}.$$

Thus,

$$ax - my = 1$$

for some integer y . Any common divisor of a and m must divide the left-hand side, and therefore must divide 1 on the right hand side. Hence,

$$\gcd(a, m) = 1.$$

Conversely, suppose

$$\gcd(a, m) = 1.$$

By Bézout's identity, there exist integers x and y such that

$$ax + my = 1.$$

Reducing modulo m gives

$$ax \equiv 1 \pmod{m}.$$

Therefore, x is an inverse of a modulo m . □

Example 11

Find the inverse of 7 modulo 12.

Since

$$7 \cdot 7 = 49 \equiv 1 \pmod{12},$$

the inverse of 7 modulo 12 is

$$\boxed{7}.$$

Example 12

Solve

$$5x \equiv 3 \pmod{11}.$$

The inverse of 5 modulo 11 is 9 because

$$5 \cdot 9 = 45 \equiv 1 \pmod{11}.$$

Multiplying both sides by 9 gives

$$x \equiv 9 \cdot 3 = 27 \equiv 5 \pmod{11}.$$

Thus,

$$\boxed{x \equiv 5 \pmod{11}}.$$

Cycles of Powers

Return to the opening question: what is the units digit of

$$7^{2026}?$$

The units digit of a number is its remainder modulo 10. Compute the first few powers:

$$7^1 \equiv 7 \pmod{10},$$

$$7^2 \equiv 9 \pmod{10},$$

$$7^3 \equiv 3 \pmod{10},$$

$$7^4 \equiv 1 \pmod{10}.$$

After reaching 1, the pattern repeats:

$$7, 9, 3, 1, 7, 9, 3, 1, \dots$$

Since the cycle has length 4 and

$$2026 \equiv 2 \pmod{4},$$

we have

$$7^{2026} \equiv 7^2 \equiv 9 \pmod{10}.$$

Therefore, the units digit is

$$\boxed{9}.$$

Possible Cycle Lengths for Units Digits

The units digits of powers depend only on the base modulo 10.

- Bases ending in 0, 1, 5, or 6 have cycle length 1.
- Bases ending in 4 or 9 have cycle length 2.
- Bases ending in 2, 3, 7, or 8 have cycle length 4.

Thus, every units-digit power cycle has length 1, 2, or 4.

Example 13

Find the units digit of

$$3^{3^3}.$$

The units digits of powers of 3 repeat every 4:

$$3, 9, 7, 1, \dots$$

Therefore, we only need the exponent

$$3^3 = 27$$

modulo 4. Since

$$27 \equiv 3 \pmod{4},$$

we have

$$3^{27} \equiv 3^3 \equiv 7 \pmod{10}.$$

Thus, the units digit is

$$\boxed{7}.$$

Fermat's Little Theorem

Power cycles become especially useful when the modulus is prime.

Theorem N5.1.14 (Fermat's Little Theorem)

Let p be prime. If

$$p \nmid a,$$

then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, for every integer a ,

$$a^p \equiv a \pmod{p}.$$

Proof. Consider the nonzero residues modulo p :

$$1, 2, \dots, p-1.$$

Since $\gcd(a, p) = 1$, multiplication by a permutes these residues. In other words,

$$a, 2a, \dots, (p-1)a$$

are congruent modulo p , in some order, to

$$1, 2, \dots, p-1.$$

(Think: Why is this so?)

Multiplying all the terms gives

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Because none of $1, 2, \dots, p-1$ is divisible by p , the number $(p-1)!$ has an inverse modulo p . Cancelling it gives

$$a^{p-1} \equiv 1 \pmod{p}.$$

□

Example 15

Find the remainder when

$$2^{1000}$$

is divided by 13.

Since 13 is prime and $13 \nmid 2$, Fermat's Little Theorem gives

$$2^{12} \equiv 1 \pmod{13}.$$

Also,

$$1000 \equiv 4 \pmod{12}.$$

Therefore,

$$2^{1000} \equiv 2^4 = 16 \equiv 3 \pmod{13}.$$

The remainder is

$$\boxed{3}.$$

Euler's Theorem

Fermat's Little Theorem applies when the modulus is prime. Euler's theorem generalizes it to composite moduli.

Definition N5.1.16 (Euler's Totient Function). For a positive integer n , the value $\varphi(n)$ is the number of integers among

$$1, 2, \dots, n$$

that are relatively prime to n .

For example, the positive integers at most 10 that are relatively prime to 10 are

$$1, 3, 7, 9,$$

so

$$\varphi(10) = 4.$$

Theorem N5.1.17 (Euler's Theorem)

If

$$\gcd(a, n) = 1,$$

then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Proof. Let

$$r_1, r_2, \dots, r_{\varphi(n)}$$

be the residues between 1 and n that are relatively prime to n .

Since $\gcd(a, n) = 1$, multiplying each r_i by a permutes this set modulo n . Therefore,

$$(ar_1)(ar_2) \cdots (ar_{\varphi(n)}) \equiv r_1 r_2 \cdots r_{\varphi(n)} \pmod{n}.$$

Thus,

$$a^{\varphi(n)} r_1 r_2 \cdots r_{\varphi(n)} \equiv r_1 r_2 \cdots r_{\varphi(n)} \pmod{n}.$$

Canceling gives

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

□

Example 18

Find the remainder when

$$3^{100}$$

is divided by 10.

Since

$$\gcd(3, 10) = 1$$

and

$$\varphi(10) = 4,$$

Euler's theorem gives

$$3^4 \equiv 1 \pmod{10}.$$

Because

$$100 \equiv 0 \pmod{4},$$

we obtain

$$3^{100} \equiv 1 \pmod{10}.$$

Thus, the remainder is

$$\boxed{1}.$$

Wilson's Theorem

Wilson's theorem gives another striking property of prime numbers.

Theorem N5.1.19 (Wilson's Theorem)

A positive integer $p > 1$ is prime if and only if

$$(p-1)! \equiv -1 \pmod{p}.$$

Proof. First suppose that p is prime. Every nonzero residue modulo p has a unique inverse. Most residues can be paired with a distinct inverse, and each such pair has product 1 modulo p .

The only residues that are their own inverses satisfy

$$x^2 \equiv 1 \pmod{p}.$$

Since p is prime,

$$(x-1)(x+1) \equiv 0 \pmod{p}$$

implies

$$x \equiv 1 \pmod{p} \quad \text{or} \quad x \equiv -1 \pmod{p}.$$

Thus, after pairing all other residues with their inverses,

$$(p-1)! \equiv 1 \cdot (-1) \equiv -1 \pmod{p}.$$

Conversely, if p is composite, then $(p-1)!$ is divisible by p except for the easily checked case $p = 4$, where

$$3! = 6 \not\equiv -1 \pmod{4}.$$

Hence a composite p cannot satisfy Wilson's congruence. □

Example 20

Find

$$10! \pmod{11}.$$

Since 11 is prime, Wilson's theorem gives

$$10! \equiv -1 \equiv 10 \pmod{11}.$$

Thus, the remainder is

$$\boxed{10}.$$

Possible Remainders of Powers

Modular arithmetic can be used not only to compute remainders, but also to prove that certain equations are impossible.

Squares Modulo 4

Every integer is congruent to one of

$$0, 1, 2, 3 \pmod{4}.$$

Squaring gives

$$0^2 \equiv 0, \quad 1^2 \equiv 1, \quad 2^2 \equiv 0, \quad 3^2 \equiv 1 \pmod{4}.$$

Therefore,

$$\boxed{n^2 \equiv 0 \text{ or } 1 \pmod{4}}.$$

Claim N5.1.21 — No integer square is congruent to 2 or 3 modulo 4.

Example 22

Prove that the equation

$$x^2 + y^2 = 4k + 3$$

has no integer solutions.

Every square is congruent to 0 or 1 modulo 4. Therefore,

$$x^2 + y^2$$

can be congruent only to

$$0, 1, \text{ or } 2 \pmod{4}.$$

It can never be congruent to 3 (mod 4). Hence the equation has no integer solutions.

Odd Squares Modulo 8

Every odd integer is congruent to one of

$$1, 3, 5, 7 \pmod{8}.$$

Squaring each gives

$$1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}.$$

Therefore,

$$\boxed{n \text{ odd} \implies n^2 \equiv 1 \pmod{8}}.$$

Squares Modulo 3

The residues modulo 3 are 0, 1, 2. Their squares are

$$0, 1, 1.$$

Thus,

$$\boxed{n^2 \equiv 0 \text{ or } 1 \pmod{3}}.$$

Cubes Modulo 9

Checking the residues $0, 1, \dots, 8$ shows that every cube is congruent to

$$0, 1, \text{ or } -1 \pmod{9}.$$

Thus,

$$\boxed{n^3 \equiv 0, \pm 1 \pmod{9}}.$$

Example 23

Show that

$$x^3 + y^3 = 3$$

has no integer solutions.

Each cube is congruent to 0, 1, or -1 modulo 9. Therefore, the sum of two cubes can only be congruent to

$$-2, -1, 0, 1, \text{ or } 2 \pmod{9}.$$

It cannot be congruent to $3 \pmod{9}$. Hence there are no integer solutions.

Fourth Powers Modulo 16

If n is even, then

$$n^4 \equiv 0 \pmod{16}.$$

If n is odd, then

$$n^2 \equiv 1 \text{ or } 9 \pmod{16},$$

and squaring again gives

$$n^4 \equiv 1 \pmod{16}.$$

Therefore,

$$\boxed{n^4 \equiv 0 \text{ or } 1 \pmod{16}}.$$

The Chinese Remainder Theorem

Sometimes we want a number to satisfy several remainder conditions at once.

Example 24

Find the least positive integer x satisfying

$$x \equiv 2 \pmod{3}$$

and

$$x \equiv 3 \pmod{5}.$$

The numbers congruent to 3 modulo 5 are

$$3, 8, 13, 18, \dots$$

The first of these congruent to 2 modulo 3 is 8. Therefore,

$$\boxed{x \equiv 8 \pmod{15}}.$$

Theorem N5.1.25 (Chinese Remainder Theorem)

If

$$\gcd(m, n) = 1,$$

then for any integers a and b , the system

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n}$$

has exactly one solution modulo mn .

Proof. Because $\gcd(m, n) = 1$, the number m has an inverse modulo n . Let m^{-1} denote such an inverse.

Write

$$x = a + mk.$$

This automatically ensures that

$$x \equiv a \pmod{m}.$$

To satisfy the second congruence, we need

$$a + mk \equiv b \pmod{n},$$

so

$$mk \equiv b - a \pmod{n}.$$

Multiplying by the inverse of m modulo n gives a unique value of k modulo n . Therefore, x is uniquely determined mod mn . $\square$

Remark N5.1.26. If m and n are not relatively prime, the system

$$x \equiv a \pmod{m}, \quad x \equiv b \pmod{n}$$

has a solution exactly when

$$a \equiv b \pmod{\gcd(m, n)}.$$

When a solution exists, it is unique modulo

$$\text{lcm}(m, n).$$

Modular Invariants

A quantity is called an **modular invariant** if its remainder modulo some number remains unchanged throughout a process.

Example 27 (Token Moves on a Board)A token begins at $(0, 0)$. On each move, it may move by one of the vectors

$$(2, 1), \quad (1, 2), \quad (-2, -1), \quad (-1, -2).$$

Can the token reach $(1, 1)$?

For every allowed move, the quantity

$$x + y$$

changes by either 3 or -3 . Therefore,

$$x + y \pmod{3}$$

is invariant.

Initially,

$$x + y = 0.$$

At the point $(1, 1)$,

$$x + y = 2.$$

Since

$$0 \not\equiv 2 \pmod{3},$$

the token can never reach $(1, 1)$.

Example 28 (Removing Tokens)

A pile begins with 100 tokens. Two players alternate removing 1, 2, 3, or 4 tokens. The player who removes the last token wins. Which player has a winning strategy?

No matter how many tokens t player 1 takes, the player 2 can always take a corresponding amount of tokens $5 - t$. Specifically:

- if the first player removes 1, the second removes 4;
- if the first player removes 2, the second removes 3;
- if the first player removes 3, the second removes 2;
- if the first player removes 4, the second removes 1.

Since

$$100 \equiv 0 \pmod{5},$$

After each second-player move, the number of remaining tokens is again a multiple of 5. Eventually, the second player removes the last token and wins.

Remark N5.1.29. If the game begins with 101, 102, 103, or 104 tokens, the first player can remove enough tokens to leave 100, then use the same complementary strategy.